

Part VIII — Electricity and Magnetism · Chapter 45

Electromagnetic Induction

How changing magnetic fields create electric currents — and how that discovery powers the modern world

Every time you plug a phone charger into a wall socket, you depend on electromagnetic induction. Every power plant on Earth — whether it burns coal, splits uranium, harnesses the wind, or catches falling water — ultimately converts mechanical energy into electrical energy through induction. Every transformer on every utility pole steps voltage up or down through induction. The discovery that a changing magnetic field can drive an electric current was made in 1831 and transformed civilisation within a generation. Michael Faraday, working with simple coils and magnets and no formal mathematical training, uncovered the operating principle of the entire electrical power industry. This chapter explains that principle: what electromagnetic induction is, why it occurs, and how engineers have used it to build generators, transformers, and every piece of infrastructure that makes electricity available on demand.

45.1 The Discovery of Electromagnetic Induction

By 1820, Hans Christian Ørsted had shown that an electric current produces a magnetic field (Chapter 44). The natural next question was whether the reverse was true: can a magnetic field produce an electric current?

For over a decade, many physicists tried to produce a current using a static magnet placed near a wire. None succeeded. The crucial insight — that the magnetic field must be changing, not merely present — was discovered independently and nearly simultaneously by Michael Faraday in England and Joseph Henry in the United States in 1831.

Historical Note: Faraday's original experiment used a ring of soft iron wound with two separate coils of wire. When he connected a battery to one coil (the primary), a current appeared briefly in the other coil (the secondary) — but only at the moment of connection or disconnection. A steady current in the primary produced no effect. It was the change in the magnetic field threading the secondary coil that drove the current. Faraday had no mathematical framework for this; he thought in terms of "lines of force" filling space. His physical intuition was so acute that he reached the correct conclusion anyway, and James Clerk Maxwell later gave it the mathematical form we use today.

Henry made a similar discovery at about the same time but published later, losing priority. He did, however, discover self-inductance (a coil inducing a voltage in itself due to its own changing current) somewhat before Faraday, and the SI unit of inductance, the henry (H), commemorates him.

The unifying idea is this: a changing magnetic environment can push electric charges into motion. Whether the magnetic field changes because a magnet moves, because a current ramps up or down, or because a conductor moves through a steady field, the result is the same — charges experience a force and current can flow if a circuit is present.

45.2 Magnetic Flux

To state the law of induction precisely, we need the concept of magnetic flux. Magnetic flux (Φ , phi) through a surface is a measure of how many magnetic field lines pass through that surface. It combines the field strength, the area of the surface, and the angle between the field lines and the surface.

$$\Phi = B \times A \times \cos(\theta)$$

Magnetic flux Φ (webers, Wb) = magnetic field B (tesla, T) $\times$ area A (m^2) $\times \cos\theta$, where θ is the angle between the magnetic field direction and the normal (perpendicular) to the surface. $1 \text{ Wb} = 1 \text{ T}\cdot\text{m}^2$.

When the field is perpendicular to the surface ($\theta = 0^\circ$), $\cos\theta = 1$ and the flux is maximum: $\Phi = BA$. When the field is parallel to the surface ($\theta = 90^\circ$), $\cos\theta = 0$ and the flux is zero — the field lines skim along the surface without passing through it. This makes physical sense: flux counts lines passing through the area, not lines running alongside it. Think of it this way: if you hold a wire loop and dip it into a pool of water, the flux is analogous to the volume of water flowing through the loop per unit time — maximum when the loop is horizontal (perpendicular to flow), zero when the loop is vertical (parallel to flow).

Situation	θ (field to surface normal)	Φ
Field perpendicular to surface	0°	BA (maximum)
Field at 45° to surface	45°	$BA/\sqrt{2} \approx 0.707 BA$
Field parallel to surface	90°	0 (minimum)
Field reversed (anti-perpendicular)	180°	$-BA$

Table 45.1 *Magnetic flux for various orientations of field relative to surface. The sign of flux depends on the chosen normal direction; only changes in flux have physical consequence.*

Key Point: Flux can be positive or negative depending on the direction of the field through the surface, but only the change in flux matters for induction. A large steady flux produces no current; a small but rapidly changing flux can produce a large current.

45.3 What Causes Induced Current

An induced current appears in a closed conducting loop whenever the magnetic flux through the loop changes. The flux can change in three distinct ways:

- The magnetic field B changes in strength while the loop stays fixed. (Example: a nearby electromagnet is switched on or off, or a current in an adjacent coil is varied.)
- The area A of the loop changes while the field stays constant. (Example: a loop is deformed, stretched, or compressed inside a uniform field.)
- The angle θ between the field and the loop changes. (Example: the loop rotates inside a uniform field — the operating principle of every electric generator.)

In each case, the flux $\Phi = BA \cos\theta$ changes over time, and that changing flux is what drives the induced current. A static situation — strong field, large loop, but nothing changing — produces no induced current at all. The current depends on the rate of change of flux, not on the flux itself.

The mechanism by which changing flux drives current is an induced electric field. A changing magnetic field creates an electric field in the surrounding space even in the absence of any conductor. When a conductor is present, the induced electric field exerts forces on the free charges in the conductor, driving a current. When no conductor is present, the induced electric field still exists — it just has nothing to

accelerate into a steady flow. This is one of Maxwell's most profound insights: changing magnetic fields and changing electric fields are not independent; each can generate the other.

45.4 Faraday's Law

Faraday's Law states the quantitative relationship between the rate of change of magnetic flux and the induced electromotive force (EMF). In conceptual form:

The Rule: The induced EMF in a loop is proportional to the rate at which the magnetic flux through the loop is changing. The faster the flux changes, the larger the induced EMF.

$$\varepsilon = -N \times (\Delta\Phi / \Delta t)$$

Faraday's Law. ε = induced EMF (volts); N = number of turns in the coil; $\Delta\Phi$ = change in magnetic flux (Wb); Δt = time interval (s). The negative sign encodes Lenz's Law (the induced current opposes the change causing it).

Several consequences follow immediately. First, a coil of N turns is N times more effective than a single loop because each turn contributes an independent EMF that adds to the others. This is why transformer coils and generator windings contain many turns of wire. Second, the induced EMF is proportional to $\Delta\Phi/\Delta t$, the rate of change of flux — not the amount of flux. Third, the negative sign tells us the direction of the induced EMF opposes the change, a constraint known as Lenz's Law (Section 45.5).

Example — Faraday's Law calculation A 200-turn coil experiences a change in magnetic flux from 0.05 Wb to 0.12 Wb in 0.20 s. $\Delta\Phi = 0.07$ Wb. $\varepsilon = N(\Delta\Phi/\Delta t) = 200 \times 0.07/0.20 = 70$ V. This EMF drives a current through whatever external circuit is connected to the coil.

Faraday's Law is one of the four Maxwell equations that govern all classical electromagnetic phenomena. It tells us that electricity and magnetism are not independent: you cannot have a changing magnetic field without a corresponding electric effect. This mutual dependence is ultimately what allows electromagnetic waves to propagate through empty space (Chapter 46).

45.5 Lenz's Law and Direction of Induced Current

Faraday's Law gives the magnitude of the induced EMF. Lenz's Law gives its direction: the induced current always flows in the direction that opposes the change in flux that caused it.

The Rule: Lenz's Law: The induced current creates a magnetic field that opposes the change in flux through the loop. If the flux is increasing, the induced current's field opposes the increase (reduces the flux). If the flux is decreasing, the induced current's field opposes the decrease (adds to the flux).

Lenz's Law is a consequence of energy conservation. If the induced current reinforced the change that caused it, the resulting magnetic field would cause more induction, which would increase the current further, which would increase the field further, producing energy from nothing. Lenz's law prevents this runaway: the induced current always costs energy to drive, and that energy must come from the agent that is changing the flux (e.g., the hand pushing a magnet, or the turbine turning a generator).

A helpful mnemonic: the induced current is always fighting the change. Push a north pole toward a loop: the loop becomes an electromagnet with its own north pole facing the approaching magnet, repelling it. Pull the north pole away: the loop becomes a south pole facing the retreating magnet, attracting it. In both cases, you must do work against the magnetic force, and that work becomes electrical energy in the circuit.

Example — Lenz's Law direction A horizontal loop lies flat. A bar magnet is held above the loop with its north pole pointing downward toward the loop. As the magnet falls closer, the downward flux through the loop increases. Lenz's Law: the induced current must create a field opposing the increase — that is, an upward field through the loop. By the right-hand rule, when the right-hand fingers curl in the direction of current flow and the thumb points upward (through the loop), the current flows counterclockwise as seen from above.

Common Mistake: Students often confuse the direction of the induced magnetic field with the direction of the induced current. First determine the required field direction (Lenz's Law), then apply the right-hand rule to find the current direction. Always do these as two separate steps.

45.6 Induced EMF

An electromotive force (EMF) is not actually a force — it is a voltage (energy per unit charge) that drives charges around a circuit. Induced EMF plays the same role as the voltage of a battery: it pushes current through the circuit. Unlike a battery, which converts chemical energy to electrical energy, an induced EMF converts mechanical energy (the work done to change the flux) to electrical energy.

The induced EMF drives a current through the loop according to Ohm's Law: $I = \mathcal{E}/R$, where R is the total resistance of the circuit. The greater the rate of flux change, the greater the EMF, and the greater the current. The greater the resistance, the smaller the current for the same EMF.

Example — Induced current A single circular loop of wire (radius 0.10 m, resistance 2.0 Ω) lies perpendicular to a magnetic field that decreases from 0.80 T to 0.20 T in 0.50 s. Area $A = \pi(0.10)^2 = 0.0314 \text{ m}^2$. $\Delta\Phi = A\Delta B = 0.0314 \times (0.80 - 0.20) = 0.0314 \times 0.60 = 0.01885 \text{ Wb}$. $\mathcal{E} = \Delta\Phi/\Delta t = 0.01885/0.50 = 0.0377 \text{ V}$. $I = \mathcal{E}/R = 0.0377/2.0 = 0.019 \text{ A} \approx 19 \text{ mA}$. By Lenz's Law, since the flux is decreasing, the current flows in the direction that tries to maintain it (right-hand rule determines the actual direction).

The power delivered by the induced EMF is $P = \mathcal{E}I = \mathcal{E}^2/R$. This power comes from the mechanical energy of whatever agent is changing the flux. An electric generator, for example, is mechanically driven by a turbine or engine, and the mechanical power delivered equals (in the ideal case) the electrical power produced. Real generators have resistive losses in the windings and friction in the bearings, reducing efficiency below 100%, but modern large generators achieve efficiencies above 98%.

45.7 Moving Conductors in Magnetic Fields

When a straight conductor of length L moves with velocity v perpendicular to a uniform magnetic field B , an EMF is induced across the conductor. This is sometimes called motional EMF. It arises because the magnetic field exerts a force on the moving positive charges in the conductor ($F = qv \times B$), pushing them to one end and creating a charge separation — a voltage.

$$\varepsilon = BLv$$

Motional EMF for a conductor of length L (m) moving at velocity v (m/s) perpendicular to field B (T). If the conductor forms part of a closed circuit, this EMF drives a current $I = \varepsilon/R$.

This is the basis for railgun technology, MHD (magnetohydrodynamic) propulsion, and some types of electric generators. In a generator, the rotating coil has conductors that sweep through the magnetic field; the motional EMF in each conductor adds up to drive the total generator output.

Note that if the conductor is not part of a closed circuit, charges accumulate at the ends until the electric force due to the charge separation exactly cancels the magnetic force. No current flows, but a voltage $\varepsilon = BLv$ exists across the ends. This is analogous to an open-circuit battery: the EMF is present but no current flows because there is no complete path.

Example — Motional EMF A 0.50 m copper rod slides along two parallel rails separated by 0.50 m in a 0.30 T magnetic field perpendicular to the rails. The rod moves at 2.0 m/s. $\varepsilon = BLv = 0.30 \times 0.50 \times 2.0 = 0.30$ V. If the rails are connected by a 1.5 Ω resistor, the current is $I = 0.30/1.5 = 0.20$ A. To maintain constant velocity, an external force must push the rod; that force does work at the rate $P = Fv = \varepsilon I = 0.30 \times 0.20 = 0.060$ W = 60 mW, which is exactly the power dissipated in the resistor.

45.8 Electric Generators

An electric generator converts mechanical energy into electrical energy by rotating a coil of wire inside a magnetic field (or equivalently, rotating a magnet inside a coil). As the coil rotates, the angle between the field and the coil's normal changes continuously, causing the flux to oscillate: $\Phi = BA \cos(\omega t)$, where ω is the angular speed in radians per second.

Because the flux changes sinusoidally, so does the induced EMF:

$$\epsilon(t) = NBA\omega \sin(\omega t) = \epsilon_0 \sin(\omega t)$$

Alternating EMF of a generator: $\epsilon_0 = NBA\omega$ is the peak EMF; N = number of turns; B = field strength (T); A = coil area (m^2); ω = angular frequency (rad/s); t = time (s). This is the origin of alternating current (AC).

The output is alternating current (AC): the current changes direction twice per revolution. For a generator spinning at 50 Hz (50 revolutions per second, as in European power systems) or 60 Hz (as in American power systems), the current reverses direction 100 or 120 times per second.

To produce direct current (DC), a device called a commutator is used in place of slip rings. The commutator reverses the connection to the external circuit every half turn, so the output voltage always has the same polarity (though it still pulsates). Practical DC generators use multiple coils offset by small angles to smooth out the pulsation.

Generator Type	Output	Mechanism	Applications
AC generator (alternator)	Sinusoidal AC	Slip rings; coil rotates in field	Power stations, car alternators
DC generator (dynamo)	Pulsating DC	Commutator reverses connection	Legacy applications; mostly replaced
Permanent magnet generator	AC or DC	Rotating magnets; fixed coils	Wind turbines, bicycle dynamos
Linear generator	AC or DC	Reciprocating (not rotating) motion	Free-piston engines, wave energy

Table 45.2 *Types of electric generators and their applications.*

The peak EMF of a generator, $\epsilon_0 = NBA\omega$, shows that output can be increased by increasing N (more turns), B (stronger field — using powerful electromagnets), A (larger coil area), or ω (faster rotation). Large power plant generators spin at 1500 or 3000 rpm (for 50 Hz systems) or 1800 or 3600 rpm (for 60 Hz systems) and are precisely controlled to maintain frequency.

45.9 Transformers

A transformer uses electromagnetic induction to change the voltage (and correspondingly the current) of an alternating electrical signal. It consists of two coils — the primary and the secondary — wound around a common iron core. The iron core greatly amplifies the magnetic flux linkage between the coils.

When an alternating current flows in the primary coil, it creates an alternating magnetic field in the iron core. That alternating field threads the secondary coil and induces an EMF in it by Faraday's Law. The ratio of the output (secondary) voltage to the input (primary) voltage equals the ratio of the number of turns:

$$V_2 / V_1 = N_2 / N_1$$

Transformer voltage ratio. V_1 = primary voltage; V_2 = secondary voltage; N_1 = number of primary turns; N_2 = number of secondary turns. A step-up transformer has $N_2 > N_1$ (output voltage exceeds input). A step-down transformer has $N_2 < N_1$.

Energy conservation requires that, in an ideal transformer (no losses), the power input equals the power output: $P_1 = P_2$, so $V_1 I_1 = V_2 I_2$. Therefore:

$$I_2 / I_1 = N_1 / N_2$$

Transformer current ratio. When voltage is stepped up, current is stepped down by the same factor, and vice versa. Power is conserved (in the ideal case).

Example — Transformer A step-up transformer has 500 primary turns and 10 000 secondary turns. Primary voltage is 240 V. Secondary voltage: $V_2 = V_1 \times N_2/N_1 = 240 \times 10000/500 = 4800$ V. If primary current is 20 A, secondary current: $I_2 = I_1 \times N_1/N_2 = 20 \times 500/10000 = 1.0$ A. Primary power: $240 \times 20 = 4800$ W. Secondary power: $4800 \times 1.0 = 4800$ W. Confirmed — power is conserved.

Key Point: Transformers work only with alternating current. A DC current creates a steady magnetic field; a steady field produces no change in flux; no change in flux means no induced EMF in the secondary. This is why the world's power grids use AC: it is the only form of electricity that can be transformed cheaply and efficiently.

45.10 Power Transmission Systems

One of the most important practical uses of transformers is efficient long-distance electrical power transmission. Power is lost as heat in transmission lines due to the resistance of the wires: $P_{\text{loss}} = I^2R$. To minimise this loss, the current I must be kept as small as possible.

Since power $P = VI$, transmitting a given amount of power at very high voltage allows the current to be very low, dramatically reducing transmission losses. Step-up transformers at power stations raise voltage from the generator output (typically 10–25 kV) to very high transmission voltages (110–765 kV in different countries). Near consumers, step-down transformers reduce the voltage to safe levels for distribution (typically 11 kV at local substations, then again to 240 V or 120 V at the pole-mounted transformers near homes).

Example — Transmission losses A power station generates 100 MW at 20 kV. A step-up transformer raises this to 400 kV for long-distance transmission. (a) Transmission current: $I = P/V = 100 \times 10^6 / 400 \times 10^3 = 250$ A. (b) If the transmission line has resistance 10 Ω , power lost: $P_{\text{loss}} = I^2R = 250^2 \times 10 = 625\,000$ W = 0.625 MW. Loss = 0.625%. (c) Without the transformer, at 20 kV: $I = 100 \times 10^6 / 20 \times 10^3 = 5000$ A. Power lost: $5000^2 \times 10 = 250$ MW — more than the power being transmitted! High-voltage transmission makes long-distance power delivery practical.

The modern electrical grid is a hierarchy of voltage transformations: generation (10–25 kV) → transmission (110–765 kV) → sub-transmission (33–66 kV) → distribution (11 kV) → service (240 V or 120 V). Each step-down uses transformers. The grid contains

millions of transformers, most of them small pole-top or pad-mounted units that serve individual buildings or blocks.

45.11 Induction in Everyday Technology

Electromagnetic induction is not confined to power stations. It underlies a remarkable range of everyday technologies.

Wireless Charging

Wireless chargers for smartphones, electric toothbrushes, and electric vehicles use inductive coupling. The charger contains a primary coil carrying alternating current at moderate frequency (typically 100–200 kHz for phone chargers, higher for industrial systems). The device contains a secondary coil. Changing current in the primary creates a changing magnetic field that induces an EMF in the secondary, which is rectified and used to charge the battery. No electrical contact is required.

Induction Cooktops

An induction cooktop contains a coil carrying high-frequency AC (20–50 kHz). When a steel or cast-iron pan is placed on the cooktop surface, the alternating magnetic field induces large eddy currents in the pan's base. These eddy currents, flowing in a resistive material, heat the pan directly by the I^2R effect. The cooktop surface itself does not heat up (it is glass-ceramic and non-magnetic); only the pan heats. Response is faster than gas or radiant electric, and the surface is safe to touch immediately after removing the pan.

Metal Detectors

A metal detector contains a transmitter coil and a receiver coil. The transmitter sends AC magnetic flux into the ground. Metal objects in the ground carry induced eddy currents, which create their own magnetic fields. The receiver coil detects the disturbance in the total field. The frequency and phase of the distortion can identify metals (ferrous vs. non-ferrous) and sometimes estimate depth.

Magnetic Braking

Moving a conducting plate through a magnetic field induces eddy currents in the plate. By Lenz's Law, these currents create forces opposing the motion — a braking force proportional to velocity. Magnetic brakes require no contact and no wear. They are

used in high-speed rail systems, roller coasters, laboratory instruments, and equipment that must be stopped smoothly and controllably without friction.

RFID and NFC

Radio-frequency identification (RFID) tags and near-field communication (NFC) chips in contactless cards and phones operate through inductive coupling. A reader creates an alternating magnetic field; the tag's antenna coil harvests energy from this field (via induction), powers the chip, and transmits data back by modulating its own inductance. Passive RFID tags require no battery: they are powered entirely by the induced EMF from the reader.

45.12 Common Mistakes in Induction Analysis

- Confusing flux and field. Flux ($\Phi = BA \cos\theta$) is not the same as the magnetic field B . A large field threading a tiny area gives the same flux as a small field threading a large area. Induced EMF depends on the change in flux, not the change in field alone.
- Thinking a large flux induces a current. Only a changing flux induces a current. A coil sitting inside a strong, steady magnetic field has zero induced EMF. Students must always ask: is the flux changing? If not, there is no induction.
- Ignoring the number of turns N . Faraday's Law is $\varepsilon = N\Delta\Phi/\Delta t$. A 1000-turn coil experiencing the same rate of flux change as a 1-turn loop produces 1000 times the EMF. Always include N .
- Getting Lenz's Law direction backwards. The induced current opposes the change, not the flux itself. If the flux is increasing, the induced field opposes the increase. If the flux is decreasing, the induced field tries to maintain it. Use two steps: (1) determine required field direction from Lenz's Law; (2) use right-hand rule to find current direction.
- Applying transformer equations to DC. Transformers work only with changing (AC) currents. A DC current produces a steady field, no change in flux, no induced EMF, no transformer action. Using the turns ratio with a DC input gives a wrong (and physically impossible) result.
- Forgetting that step-up voltage means step-down current. Power conservation: $V_1 I_1 = V_2 I_2$ (ideal transformer). When voltage doubles, current halves. Students sometimes expect both voltage and current to increase, which would violate energy conservation.

- Confusing EMF with current. EMF is the driving voltage (ϵ in volts). Current is $I = \epsilon/R$. A large EMF in a high-resistance circuit produces a small current. A small EMF in a low-resistance circuit can produce a large current. Always specify which quantity is being calculated.

Common Mistake: The most common error in transformer problems is applying the voltage ratio and ignoring current. Always check using $P_1 = P_2$. If the calculated secondary power exceeds the primary power, you have made an error in sign or ratio direction.

Chapter Summary

Electromagnetic induction is the generation of an EMF by a changing magnetic flux. It is the operating principle of every generator, transformer, and wireless power transfer system.

- Faraday's discovery (1831): changing flux $\rightarrow$ induced EMF $\rightarrow$ induced current. Static flux produces no current.
 - Magnetic flux: $\Phi = BA \cos\theta$ (Wb). Flux depends on field strength, area, and angle.
 - Faraday's Law: $\varepsilon = N\Delta\Phi/\Delta t$. EMF proportional to rate of flux change and number of turns.
 - Lenz's Law: induced current opposes the change in flux. Required by energy conservation.
 - Motional EMF: $\varepsilon = BLv$ for a conductor of length L moving at velocity v perpendicular to field B .
 - AC generator: rotating coil in field produces sinusoidal output $\varepsilon(t) = NBA\omega \sin(\omega t)$.
 - Transformer: $V_2/V_1 = N_2/N_1$; $I_2/I_1 = N_1/N_2$. Ideal: power in = power out. Requires AC.
 - High-voltage transmission minimises I^2R losses over long distances.
 - Induction underlies wireless charging, induction cooktops, metal detectors, magnetic braking, and RFID.
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Key Terms

electromagnetic induction: The generation of an EMF (and resulting current) in a conducting loop by a changing magnetic flux through the loop.

magnetic flux (Φ): $\Phi = BA \cos\theta$; the measure of magnetic field lines passing through a surface; SI unit weber ($\text{Wb} = \text{T}\cdot\text{m}^2$).

Faraday's Law: $\varepsilon = N\Delta\Phi/\Delta t$; the induced EMF in a coil equals the number of turns times the rate of change of magnetic flux.

Lenz's Law: The induced current flows in the direction that opposes the change in flux that caused it; a consequence of energy conservation.

induced EMF (ε): The electromotive force (voltage) generated in a conductor by a changing magnetic flux; drives current if a closed circuit is present.

motional EMF: $\varepsilon = BLv$; the EMF generated in a straight conductor of length L moving at velocity v perpendicular to magnetic field B .

generator (alternator): A device that converts mechanical energy to electrical energy by rotating a coil in a magnetic field; produces AC.

transformer: A device using mutual induction between two coils on a common core to change AC voltage; $V_2/V_1 = N_2/N_1$.

step-up transformer: A transformer with $N_2 > N_1$; output voltage exceeds input voltage; output current is less than input current.

step-down transformer: A transformer with $N_2 < N_1$; output voltage is less than input voltage; output current exceeds input current.

eddy currents: Circulating induced currents in bulk conducting materials; produce heating (useful in induction cooktops) and braking forces.

self-inductance: The property of a coil by which a changing current through it induces an EMF in itself opposing the current change; unit: henry (H).

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain why Faraday's early attempts to produce current with a static magnet failed. What was missing from his initial setup?
 2. A flat circular loop lies perpendicular to a uniform magnetic field. Describe three different ways to change the magnetic flux through the loop without changing the field strength.
 3. A coil sits in a uniform magnetic field. The field is very strong — 10 T. Is there an induced current? Explain why or why not.
 4. State Lenz's Law in your own words and explain why it must be true as a consequence of energy conservation. What would happen if the induced current reinforced the change instead of opposing it?
 5. A bar magnet is dropped through a vertical copper tube. The magnet falls more slowly than a non-magnetic object dropped through the same tube. Explain this using Lenz's Law.
 6. Why does a transformer work only with AC and not with DC? Describe what happens when DC is applied to the primary of a transformer.
 7. Explain why high-voltage transmission lines reduce energy losses in long-distance power delivery. If resistance doubles and voltage is halved, by what factor does the power loss change?
 8. An induction cooktop does not get hot when operating without a pan, but the pan placed on it heats rapidly. Explain why, using electromagnetic induction.
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Induction Concept Problems

9. A square loop (side 0.20 m) lies perpendicular to a magnetic field. The field increases from 0 to 0.60 T in 0.30 s. (a) Find the rate of change of flux. (b) Find the induced EMF. (c) If the loop has resistance 4.0 Ω , find the induced current.
 10. A 500-turn coil has area 0.040 m². The magnetic field through the coil decreases from 1.2 T to 0.4 T in 0.10 s. Find the induced EMF.
 11. A circular loop (radius 0.15 m, resistance 3.0 Ω) rotates from 0° to 90° relative to a 0.50 T field in 0.20 s. (a) Find the initial and final flux. (b) Find the average induced EMF. (c) Find the induced current.
 12. A 0.80 m conducting rod slides along frictionless rails separated by 0.80 m in a 0.25 T field perpendicular to the rails. A 2.0 Ω resistor connects the rails. (a) If the rod moves at 1.5 m/s, find the motional EMF. (b) Find the current through the resistor. (c) Find the force needed to maintain constant velocity.
 13. A magnet moves toward a 200-turn coil and induces an EMF of 12 V. The magnet then moves at half the speed. What is the new induced EMF? Explain using Faraday's Law.
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Generator and Transformer Analysis

14. A generator coil has 80 turns, area 0.025 m², and rotates at 60 Hz in a 0.40 T field. (a) Find the angular frequency ω . (b) Find the peak EMF. (c) Write the expression $\varepsilon(t)$ for the output voltage.
15. A transformer has a primary with 240 turns connected to 120 V AC. (a) How many secondary turns are needed to produce 480 V output? (b) If the secondary current is 2.0 A, what is the primary current? (c) What is the power throughput?
16. A step-down transformer converts 11 000 V to 230 V for home distribution. (a) Find the turns ratio N_2/N_1 . (b) If a household draws 50 A at 230 V, what current flows in the primary winding? (c) What power is being delivered?
17. A power plant generates 50 MW at 15 kV. The electricity is stepped up to 275 kV for transmission along lines with total resistance 8.0 Ω . (a) Find the transmission current. (b) Find the power lost in the lines. (c) Express the loss as a percentage of generated power.

Lenz's Law Direction Exercises

For each scenario, state the direction of the induced current (clockwise or counterclockwise as seen from the specified viewpoint) and justify your answer using Lenz's Law.

18. A loop lies horizontally. A north pole is moved downward toward the top of the loop from above. (Viewed from above.)
 19. A loop lies horizontally. A south pole is moved downward toward the top of the loop from above. (Viewed from above.)
 20. A north pole is moved away (upward) from a horizontal loop below it. (Viewed from above.)
 21. A vertical loop is moved to the right into a region of uniform magnetic field directed to the right. (Viewed from the right, i.e., looking in the direction the field points.)
 22. A vertical loop sits inside a uniform field. The field strength decreases. (Viewed from the direction the original field points.)
 23. A conducting square loop is pulled out of a uniform field region that passes through the right half of the loop. The right half is inside the field; the left half is outside. The loop is pulled to the left. (Viewed from the direction the field points.)
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Applied Electromagnetic Systems Problems

24. Design problem: A generator must produce a peak EMF of 240 V at 50 Hz. The available magnetic field is 0.30 T. If the coil has 150 turns, what area must the coil have? What is the minimum coil radius if the coil is circular?
25. A wireless charger operates at 150 kHz. The primary coil has 20 turns and carries 1.5 A. The secondary coil has 40 turns. (a) What is the turns ratio? (b) If the coupling between the coils is 80% efficient (i.e., 80% of primary flux links the secondary), what is the effective secondary EMF relative to the primary EMF? (Assume ideal transformer ratio applies to the linked flux only.)
26. A magnetic braking system uses a 0.40 T field and an aluminium disk (effective resistance for eddy currents $0.010\ \Omega$). The disk moves at 5.0 m/s and has an effective conductor length of 0.10 m in the field. (a) Estimate the induced EMF. (b) Estimate the eddy current. (c) Estimate the braking force $F = BIL$. (Note: real eddy current calculations are complex; this is a simplified model.)
27. An RFID tag must harvest at least 0.50 mW from a reader's field to power its chip. The tag's coil has 8 turns and area $5.0\ \text{cm}^2$. The reader produces a field that alternates at 13.56 MHz with amplitude $1.5 \times 10^{-5}\ \text{T}$. (a) Find the maximum flux through the tag coil. (b) Estimate the peak induced EMF using $\varepsilon = N\Phi(2\pi f)$ for sinusoidal flux. (c) If the tag's load resistance is $50\ \Omega$, find the induced current and power. Is the 0.50 mW threshold met?

Chapter 45 Checkpoint

Before moving on to Chapter 46, confirm you can do each of the following.

I can...	See Section
Describe Faraday's discovery and what distinguishes a changing field from a static field	45.1
Define magnetic flux and calculate $\Phi = BA \cos\theta$	45.2
Identify the three ways magnetic flux through a loop can change	45.3
Apply Faraday's Law to calculate induced EMF from rate of flux change	45.4
Apply Lenz's Law to determine the direction of induced current	45.5
Calculate induced current using $I = \varepsilon/R$ and find power delivered	45.6
Calculate motional EMF using $\varepsilon = BLv$	45.7
Describe how AC generators work and write the output EMF equation	45.8
Apply transformer voltage and current ratios; verify power conservation	45.9
Explain why high voltage reduces transmission losses	45.10
Identify and explain induction in everyday technologies	45.11
Avoid common mistakes in direction, turns, and transformer calculations	45.12

Continue to Chapter 46: Electromagnetic Waves →